Interval notation

1

a $(-1,2]$

b $(-6,-3)$

c $[-3,\infty)$

d $(-3,\infty)$

e $(-\infty,4)$

2

a $(\infty,-7)$

b $(-\infty,6]$

c $[-4,\infty)$

d $[-7,3]$

Domain and range

3

a

i Domain$=8,1,3,9,2$

ii Range$=5,3,4,1,9$

iii Function

b

i Domain$=2,3,6,8$

ii Range$=3,8,1,7,2$

iii Not a function

c

i Domain$=4,8,2,7,3$

ii Range$=4,8,9,1$

iii Function

d

i Domain$=1,6,3,8,2$

ii Range$=3,2,7,1$

iii Function

e

i Domain$=7,8,5,3$

ii Range$=1,9,3,2,6$

iii Not a function

f

i Domain$=2,4,6,8,10$

ii Range$=2,4,6,8,9$

iii Function

g

i Domain $=3\leq x\leq 7$

ii Range $=5\leq y\leq 8$

iii Function

h

i Domain $=2\leq x\leq 7$

ii Range $=5$

iii Function

i

i Domain $=5$

ii Range $=2\leq y\leq 6$

iii Not a function

j

i Domain $=2\leq x\leq 9$

ii Range $=3\leq y\leq 8$

iii Not a function

k

i Domain $=-7\leq x\leq 7$

ii Range $=0\leq y\leq 7$

iii Function

l

i Domain $=-9\leq x\leq 9$

ii Range $=-9\leq y\leq 9$

iii Not a function

m

i All real x

ii $y \geq 0$

iii Function

i $x \geq 0$

ii All real y

iii Not a function

4

a

i Domain $= (-\infty, \infty)$

ii Range $= (-\infty, 2]$

b

i Domain $= (-\infty, \infty)$

ii Range $= [-2, \infty)$

c

i Domain$= [0, 6]$

ii Range$= [0, 7]$

5

a $(-\infty, \infty)$ **b** $(-\infty, \infty)$ **c** $[-6, 6]$

6

a False **b** False **c** True **d** True

7

a 1 **b** No **c** All real x **d** $y > 0$

e $y = 128$ **f** $x = 8$

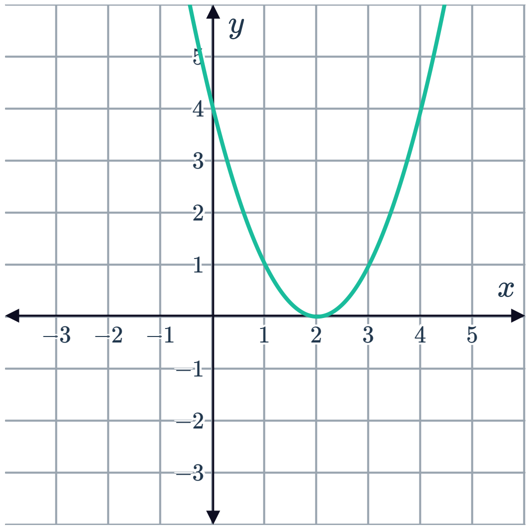
8

a 16 **b** $y \leq 16$ **c** All real x

9

a $x \geq -1$ **b** No

10
a



b No

Continuity

11
a Yes

b
i Yes ii Yes iii No

12
a Yes

b Yes

13
a Not continuous b Continuous c Not continuous d Continuous
e Continuous f Not continuous g Continuous h Not continuous

14
a False b True c False

15
a Yes b Yes c Yes

16
a Yes

b Yes

17

a $(-\infty, 2) \cup (2, \infty)$

b Yes

c Yes

18

a $(0, \infty)$

b Yes

c Yes

19

a No

b $(-\infty, 2) \cup (2, \infty)$

20

a $[0, \infty)$

b $[0, \infty)$

c Yes

21

a $(-\infty, 5) \cup (5, \infty)$

b $(-\infty, 5) \cup (5, \infty)$

c Yes

22

a $(-\infty, \infty)$

b $(-\infty, \infty)$

c Yes

23

a $(-\infty, \infty)$

b $(-\infty, -5) \cup (-5, \infty)$

c No. The function is not continuous at every point in its domain.

24

a $(-\infty, \infty)$

b $(-\infty, 1) \cup (1, \infty)$

c No. The function is discontinuous at $x = 1$.

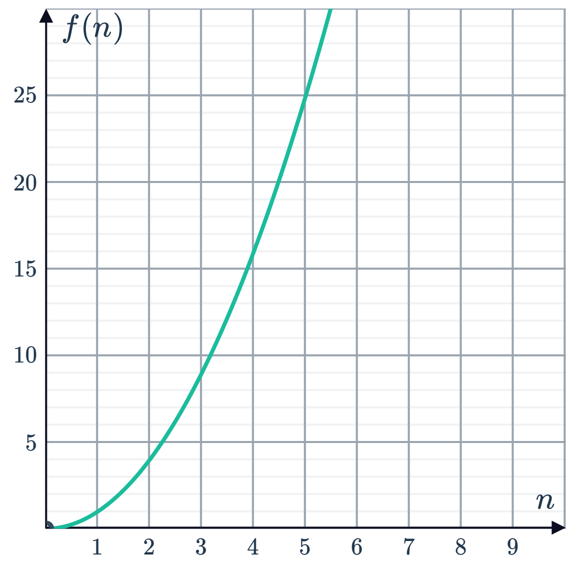
Applications

25

a i No ii Yes iii Yes iv Yes

b $f(n) = n^2$

c



26

a $0 \leq x \leq 7$

b 30

c No

27

a Yes

b The height of the water is a continuous function.

c We don't know because we don't know the location of the discontinuity.